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DUCT OR VESSEL COMPRISING A SEPARATING ELEMENT

Abstract

A duct or vessel has a separating element having a pair of plates, each of which has a central portion and a peripheral edge raised with respect to the central portion. The plates are joined to each other at their central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates. Each of the plates has a profile in radial cross-section having an inflection point between the central portion and the peripheral edge. A tangent to the plate at the inflection point defines an opening angle α with respect to an interface plane between the plates. The opening angle α satisfies the relation $\theta+\alpha\geq 90°$, where θ is a wetting angle between a molten braze-welding filler material and a base material of the separating element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Italian Patent Application No. 102024000005080 filed Mar. 7, 2024, the contents of which is hereby incorporated in its entirety by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to a duct or vessel comprising a separating element that divides the interior of the duct or vessel into a first inner space and a second inner space, and is joined to the duct or vessel by braze-welding, wherein the separating element comprises a pair of plates, each of which comprises a central portion and a peripheral edge raised with respect to the central portion, the plates being joined to each other at their respective central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates, the annular gap fluidically communicating with a through-hole formed on a side wall of the duct or vessel, wherein each of the plates has a profile in radial cross-section comprising an inflection point between the central portion and the peripheral edge, and wherein the tangent to the plate at the inflection point defines an opening angle α with respect to an interface plane between the plates.

BACKGROUND OF THE INVENTION

[0003] Ducts of this type are, for example, the distributors of heat exchangers used in HVAC systems with heat pumps.

[0004] In such heat exchangers, a separating baffle is installed to divide the passages inside the heat exchanger. When there is an internal leak from the separator (i.e., leakage between the two internal spaces created within the partitioned distributor), this results in a significant loss of performance. Detecting such leaks during the manufacturing process is highly complex, and the available detection systems often lack the sensitivity required to detect small leaks that could still negatively impact the performance of the heat exchanger. JP 2016099097 A describes a heat exchanger provided with manifolds formed by a plate and a tank assembled together and comprising a separator formed by a pair of plates provided with a protruding structure through a wall of the tank. The plates are joined to each other at their respective central portions in such a way as to define an annular gap around the central portions, interposed between the peripheral edges of the plates. The separator improves detection by ensuring that any leakage is directed towards the exterior of the heat exchanger, making it detectable through a simple pressure-tightness test.

[0005] It has been observed that the braze-welding process can cause the annular gap to become filled with the molten filler material, thereby nullifying the above-described function of the separating element.

SUMMARY OF THE INVENTION

[0006] An object of the present invention is to provide a solution capable of at least partially overcoming the aforementioned drawback.

[0007] This object is achieved, according to the present invention, by a duct or vessel of the type defined above, wherein the opening angle α satisfies the relation:

$$\theta+\alpha\geq 90°,$$

where θ is the wetting angle between the molten braze-welding filler material and the base material of the separating element.

[0008] In this way, a geometry is created that prevents the annular gap between the plates from being filled with the molten filler material during the braze-welding process. The corner fillet between the plates has a capillary pressure corresponding to its shape. Increasing the angle α results

in an increase in the capillary pressure of the fillet. Specifically, if θ+α≥90°, the shape of the fillet changes from concave to convex. At this point, the capillary pressure of the fillet transitions from negative to positive, causing the molten filler material to flow into areas with lower capillary pressure, thereby preventing the filling of the space between the two plates.
[0009] Although the present invention has been conceived within the context of separators used in the heat exchangers distributors, it is evident to a skilled person in the field that it may find application in other areas where it is necessary to prevent clogging of the annular gap defined by the separator.
[0010] Preferred embodiments of the present invention are also described.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0011] Further features and advantages of the present invention will become clearer from the following detailed description of an embodiment of the invention, made with reference to the accompanying drawings, provided purely for illustrative and non-limiting purposes, wherein:
[0012] FIG. **1** is a perspective view of a heat exchanger;
[0013] FIG. **2** is a perspective view of a detail of a duct or vessel according to the present invention;
[0014] FIG. **3** is a sectional perspective view of the detail of FIG. **2**;
[0015] FIG. **4** is a sectional view of the detail of FIG. **2**;
[0016] FIG. **5** is a geometric diagram representing a separating element of the duct or vessel of FIG. **2**;
[0017] FIG. **6** is a graph representing the capillary pressure as a function of the fillet volume for different values of the opening angle α; and
[0018] FIGS. **7** and **8** are photographs of a detail of the separating element, respectively with an opening angle α of 45° and 75°.
DETAILED DESCRIPTION
[0019] With reference to FIG. **1**, a heat exchanger made of metallic material, particularly aluminium, is generally indicated by **1**. In the illustrated example, the heat exchanger **1** is a fluid-air heat exchanger, specifically an internal condenser of an HVAC system with a heat pump for an electric vehicle. However, this example is not to be considered binding, as the present invention may find application in other fields where a separator is positioned in a duct or vessel.
[0020] The heat exchanger **1** comprises at least a first distributor **11** and at least a second distributor **12**, and a plurality of parallel, coplanar tubes **13** interconnecting the first distributor **11** and the second distributor **12**. Between adjacent tubes **13**, fins **14** are interposed, though not shown in detail. On opposite sides of the assembly formed by the tubes **13**, two side plates **15** and **16** are attached.
[0021] The above-described elements are joined together in a known manner, for example, by braze-welding.
[0022] Each of the distributors **11**, **12** has a tubular body, for example, with a circular cross-section, on a side wall **11***a*, **12***a* of which a plurality of slits **17** are formed, each of which houses an end of a respective tube **13**.
[0023] As shown in FIGS. **1** to **4**, inside at least one of the distributors **11**, **12**, a separating element **30** is arranged, which divides the interior of the partitioned distributor **11** into a first inner space **11′** and a second inner space **11″**. The separating element **30** is inserted into a linear through-hole **18**, formed on the side wall **11***a* of the partitioned distributor **11** and parallel to the slits **17** for the tubes **13**. The separating element **30** has an edge abutting against an internal surface of the partitioned distributor **11**, and opposite faces contiguous to respective edges **18′** of the through-hole **18**. The

separating element **30** is joined to the partitioned distributor **11** by material bonding, specifically by braze-welding.
[0024] The separating element **30** comprises a pair of plates **31**, **32**, each of which includes a central portion **31***a*, **32***a* and a peripheral edge **31***b*, **32***b* raised with respect to the central portion **31***a*, **32***a*. As can be appreciated in FIGS. **3-4**, the plates **31**, **32** are joined to each other at their respective central portions **31***a*, **32***a* so as to define an annular gap G around the central portions **31***a*, **32***a*. The annular gap G is interposed between the peripheral edges **31***b*, **32***b* of the plates **31**, **32**, and therefore communicates with the external environment through the part of the through-hole **18** located between the peripheral edges **31***b*, **32***b* of the plates **31**, **32**. Any fluid leakage between the edges of the separating element **30** and the internal surface of the partitioned distributor **11** is thus detectable through the through-hole **18** via a leak test.
[0025] With reference also to FIG. **5**, each of the plates **31**, **32** has a profile in radial cross-section comprising an inflection point P**1**, P**2** between the central portion **31***a*, **32***a* and the peripheral edge **31***b*, **32***b* (the arrangement of the plates **31**, **32** is symmetrical with respect to the interface plane defined between the plates **31**, **32**). The tangent to the plate **31**, **32** at the inflection point P**1**, P**2** defines an opening angle α with respect to an interface plane M between the plates **31**, **32** (for simplicity, in FIG. **5**, only the opening angle associated with plate **32** is shown, assuming that the opening angle associated with the other plate **31** is substantially the same).
[0026] The opening angle α satisfies the relation:
[00001] $+ \geq 90^\circ$,
where θ is the wetting angle between the molten braze-welding filler material and the base material of the separating element **30**.
[0027] In this way, a geometry is created that prevents the annular gap G between the plates **31**, **32** from being filled with the molten filler material during the braze-welding process. The corner fillet between the plates **31**, **32** has a capillary pressure corresponding to its shape. Increasing the angle α causes an increase in the capillary pressure of the fillet, as represented in the graph of FIG. **6**. Specifically, if $\theta+\alpha\geq 90^\circ$, the shape of the fillet changes from concave to convex. At this point, the capillary pressure of the fillet transitions from negative to positive, causing the molten filler material to flow into areas with lower capillary pressure, thereby preventing the filling of the space between the two plates. FIGS. **7** and **8** are photographs of a detail of the separating element **30** between the plates **31**, **32** and the wall **11***a* of the distributor **11**, respectively with an opening angle α of 45° and 75°. As can be observed, in the case where α=45°, the annular gap between the plates **31**, **32** is clogged by the filler material, whereas in the case where α=75°, the annular gap is substantially free from the filler material.

Claims

1. A duct or vessel comprising a separating element that divides an interior of the duct or vessel into a first inner space and a second inner space, and is joined to the duct or vessel by braze-welding, wherein the separating element comprises a pair of plates, each of which comprises a central portion and a peripheral edge raised with respect to the central portion, the plates being joined to each other at their central portions so as to define an annular gap around the central portions, interposed between the peripheral edges of the plates, the annular gap fluidically communicating with a through-hole formed on a side wall of the duct or vessel, wherein each of the plates has a profile in radial cross-section comprising an inflection point between the central portion and the peripheral edge, wherein a tangent to the plate at the inflection point defines an opening angle α with respect to an interface plane between the plates, and wherein the opening angle α satisfies the relation: $+ \geq 90^\circ$, where θ is a wetting angle between a molten braze-welding filler material and a base material of the separating element.
2. A heat exchanger comprising a first distributor and a second distributor, and a plurality of

parallel, coplanar tubes interconnecting the first distributor and the second distributor, wherein each of the first and second distributors has a tubular body, on a side wall of which a plurality of slits are formed, wherein in each slit of the plurality of slits an end of a respective tube of the plurality of parallel, coplanar tubes is inserted, and wherein at least one of the first and second distributors is the duct or vessel of claim 1.